

A 4B Language Model Without Additional Training Did Not Improve on a Rule for Simulated PLC Lockout Causes (40 vs 90 Correct of 165 the Rule Could Not Resolve); a Maintenance Rule Using the Label Boundaries Agreed With the Labels (51 of 53 Runs)

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Released 2026-10-08. Numbers are from the single official run (set generated 2026-10-06 13:54–17:02 KST; rule tables built 15:54 KST; items, model and scoring 2026-10-06 17:06 – 2026-10-07 15:28 KST; independent re-run and ladder replay of the set 2026-10-08 07:37–07:43 KST, after scoring). The pre-registration's SHA-256 was committed before the run to a private internal repository only (commit 7650685, 2026-10-06 13:53:45 KST), and the decision to publish was made on 2026-10-08, after the results were known (Section 7). Not peer-reviewed; reviews were done by AI agents, not by human reviewers (Section 10).

Abstract

A programmable logic controller (PLC) stops a machine and shows a fault code, but one code can have several causes. We asked whether a general language model, reading the event and I/O log of a simulated test rig, adds anything to a simple rule when recommending a lockout cause (T1), the next recovery step (T2), a maintenance level (T3) or the alarm to handle first (T4). The model (M1) was Qwen 3.5 4B with no additional training, answering with one letter; we read the probabilities of the choice letters at the first output token. The rule baseline (R0) maps the PLC's first fault code to candidate causes. We pre-registered four verdict rules and four predictions, fixed 26 input files by SHA-256 and ran one official set of 2,997 simulated runs (simulation; physical values assumed; no comparison with physical equipment). The pre-registration's hash was recorded before the run in a private repository only, and the decision to publish was made after the results (Section 7). The run departed from the pre-registered procedure in several places (Section 6): the replay check of the generated set, meant to come before the rules and the model, was run only after scoring (0 mismatches); the two-generation comparison passed only after one retry; the last change to the model's fixed inputs was not reviewed; and a cross-machine check was not done. **T1: both below the line.** The model's input had the fault codes removed, while R0 reads the first fault code. On the 165 cell-B lockouts whose first fault code left more than one candidate cause, R0 was right 90 times (0.545) and the rule-plus-model composite 40 times (0.242; difference -0.303 , 95% interval -0.446 to -0.172): the rule was below 0.90 and the composite did not add 0.10. **T3: the rule is enough** — R0-T3 flagged 51 of 53 maintenance-needed runs with 0 false alarms in 108 runs; R0-T3 uses the same boundaries that defined the labels, so this shows agreement between rule and labels, not maintenance skill, and under the pre-registered tie rule the verdict could not depend on the model. **T2 and T4: consistency [PASS]** (R0 7,323/7,323 and 1,374/1,383; 0 dangerous wrong answers). Three of four predictions were correct (two of them, T2/T4 and T3, close to certain by construction); prediction 1 ($R0 \geq 0.95$ on resolved lockouts) was wrong at $700/741 = 0.945$. Changing only the order in which the choices were listed changed the model's top answer on 8,404 of its 10,387 test items (81%; 228 of the 10,387 had a single choice and could not change). The design was changed twice before the official run, after an AI cross-review found that the first version's T1 verdict was decided in advance. The result applies to one untrained 4B model and one prompting method, on simulated data.

1. Question

The PLC's ladder logic and safety functions stay as they are. On top of them, an advisory model that only reads the log of the seconds before or after a lockout makes a **recommendation**; it writes nothing to the PLC. The pre-registered question is: how often does that recommendation agree with the simulated rig's ground truth, which tasks a rule alone already settles under the Section 4 lines, and what changes when the model is added?

2. Why this is a simulation-only question

Faults, wear and operator actions are injected into a simulated rig whose PLC logic is run scan by scan, so every lockout has a known injected cause and every wear run has a known level. That ground truth does not exist on a physical line. The price is that all physical values in the rig are assumptions and no result was compared with physical equipment. This report says nothing about logs from a real plant (Section 7).

3. Experimental design

3.1 Rig, faults and labels

The rig has three cells: cell A, cell B (a belt conveyor with an inspection device, an arrival sensor and a pneumatic pusher with a rod brake that diverts boxes into a chute) and a packaging cell P. A generator injects single faults and combinations of two or three faults, plus slow wear, and records PLC events, inputs (%I), outputs (%Q) and HMI words for each run. The cause label of a lockout (`cause.root_class`) is decided by re-running the run with each injected fault removed in turn and by a fault-free baseline run. Removed before items are made: lockouts whose cause cannot be isolated to one fault; lockouts that still occur when any single injected fault is removed; lockouts that also occur in the fault-free baseline run; repeated lockouts with the same first code; inputs identical to one in an earlier split; and identical inputs with conflicting cause labels. Removed T1 windows in the official set [pre-registered count by kind]: repeated same code 133, fault-free baseline lockout 68, not isolable 109, still occurring with any single fault removed 28, same input as an earlier split 103, same input with conflicting labels 2 (T2: same input as an earlier split 187; T4: repeated same code 133, same input as an earlier split 104). The pre-registration also asked for these counts by split and category; that breakdown was not made (Section 7).

Cause categories in cell B (eight choices): supply-pressure loss, flow reduction, mechanical jam, brake, belt drive, sensor, inspection device, safety intervention. Cell A has two (sensor, safety intervention) and cell P two (safety intervention, packaging device). In this catalogue "safety intervention" means a safety function that worked as designed (emergency stop pressed, light curtain entered, packaging safety permit released); no fault of a safety device itself is injected.

Splits: train, val, test_seen and **test_unseen** (two- and three-fault combinations held out by combination key). The official set used a new master seed; the pilot set was used only to draft prompts and choices and was not reused. Official set: 60 injected runs per cell × split × cause category; wear runs: 26 repetitions per single wear channel in test_seen and 80 runs of held-out wear combinations (configuration keys `single_reps.test_seen`, `unseen_runs`); **2,997 runs, 20,016 windows, 127,453 files**. The generator also produced 75,405 record-noise variants (shifted, duplicated, missing or reordered lines); as pre-registered, they are counted only and were not scored.

3.2 Tasks

Task	Input window	Ground truth	Question to the model
T1 lockout	10 s before the lockout; masked (fault words D640, D641, D647, D648 and fault-code events	Injected cause category	One letter among the

Task	Input window	Ground truth	Question to the model
cause	removed)		cell's categories
T2 recover y step	10 s after the lockout, with D641 (current recovery step shown on the HMI); fault words and fault-code events removed	Next step in the recovery table	One letter among steps 0–8
T3 mainte nance sign	300 s, cut before the run's first PLC warning or lockout; only windows with no later warning or lockout, or with one at least 5 s later (t3 . l ead_s)	Injected wear level: normal, observe, maintenance needed (stroke time 1.08× / 1.15×; belt speed deviation 2% / 5%; chatter that passes the PLC input filter)	One letter among three levels
T4 alarm priority	Events within a 2 s group; masked	First event under a fixed priority table (safety > equipment protection > quality > production > information)	One letter among the group's events

T2 and T4 are not model comparisons in this study: T2 re-reads a value the HMI already shows, and T4's answer is the priority table itself (Section 3.6). Only R0 consistency and dangerous wrong answers decide them. T1 is judged on cell B only, because in the pilot the cell alone fixed the cause in cells A (29/29) and P (48/48).

3.3 Methods

Meth od	Definition
R0- full (T1)	Uses the unmasked first PLC fault code (D640). Code → candidate categories from a fixed expected-code table. One candidate: "rule resolved". Several: "rule unresolved", answer = the candidate most frequent in train
R0- mask ed (T1, refer ence)	Event-name → category mapping on the masked input
R0 (T2, T4)	Recovery table; priority table (falls back to R1 where the table gives no order)
R0- T3	Stroke ratio ≥ 1.15 , or belt deviation $\geq 5\%$, or chatter passing the filter → maintenance needed; $\geq 1.08 / \geq 2\% \rightarrow$ observe. These are the same boundaries that made the labels
R1	Most frequent train category
M1	Qwen 3.5 4B (Qwen/Qwen3.5-4B commit 851bf6e8, Apache-2.0), converted to BF16 GGUF with llama.cpp 7049ff0 (SHA-256 7d9e8d79...c79ccc00), no additional training, thinking mode off. Prompt order: log → question → choices → "answer with one letter". At the first output token, the probabilities of the choice letters among the top 100 tokens are renormalized. Each item is asked with min(5, n!) different orders of its n choices (fixed seed e5-m1-per m-v1; the first is the original order): of the 10,387 test items, 9,501 had five orders, 658 (two choices) had two and 228 (one choice) had one. The answer is the top category of the mean over orders. An item whose top category changes across orders is flagged for human review ("human check") and stays in the denominator
Com posit e C (T1)	R0's answer on rule-resolved items, M1's answer (masked input, choosing among all of the cell's categories) on rule-unresolved items; scores are never mixed within an item

Instructions and choices were written in Korean. Logs longer than 120,000 tokens lost their oldest lines first, with no omission marker, and R0-T3 read the same truncated text as the model.

3.4 Measures

- T1 primary: top-1 accuracy on cell-B rule-unresolved items, test_seen + test_unseen together; test_unseen is also reported alone. Secondary: recall by category for four risk categories named by the domain review (Section 3.6) — safety intervention, brake, supply-pressure loss, mechanical jam; denominators below 20 are given as counts only. Confusion tables by split, the list of missed cases, test_seen items by number of injected faults.
- T3: recall and false-alarm rate per run. A positive unit is a run with at least one maintenance-needed window (a hit if any of its windows is flagged); a negative unit is a run with only negative windows. Flags on negative windows of mixed runs are counted as early alarms and not used.
- M1's T3 threshold is chosen on val as the highest value giving val recall ≥ 0.97 .
- 95% intervals: bootstrap, 10,000 resamples (seed string e5-score-boot-v1, percentiles), clustered by combination key (T1, T2, T4) or by run (T3); Clopper–Pearson where a proportion is 0 or 1. Each measure draws its own resamples (the seed string is combined with the measure's name), so two measures with the same count can have slightly different intervals (for example C and M1 at 40/165). Ratios are compared as integers.
- Dangerous wrong answer: T4 = the true first event is safety-class and another is placed first; T2 = a step the recovery table cannot reach from the current step, or step 3 (finishing push, which the PLC rejects) when the answer is 4.

3.5 Runs, fixed inputs and invalidation

All runs were on one machine (NVIDIA GB10, aarch64, 20 cores, 121 GB unified memory, Ubuntu, NVIDIA driver 580.178.04, CUDA 13.0 as reported by `nvidia-smi` on 2026-10-08). The pre-registration fixes 26 inputs by SHA-256, including catalogues, schemas, the official set configuration, prompts, model record, rule code, scorer and verdict constants (`judge.json`); the scorer reads its lines only from `judge.json`. Rule tables were rebuilt from the official train split and their SHA-256 recorded before any test file was opened; the scorer refuses to score test items without that record. For this reason three released tables (`rules_r1.json`, `rules_t1_full.json`, `rules_t1_masked.json`) have SHA-256 values different from the pre-registration's Section 12 table, which lists the tables at freezing time: as Section 12-③ requires, they were rebuilt from the official train split and their hashes recorded before test was opened (rule-table record 2026-10-06T06:54:39Z, private commit 0e0e3f6). `rules_t3.json` is unchanged.

A run is invalid if any fixed input hash differs, if the two generations of the set on the same machine differ, or if M1 fails (no choice letter in any order) on more than 5% of the judged items. At most two retries per cause. The pre-registered order (its Section 9) is: fixed-input check → two generations → independent re-run and ladder replay of the set → rules → model → scoring. Section 6 lists where the actual order differed.

3.6 Design history (all before the official run)

The pre-registration changed twice. All three versions are in the release (fixed/).

Version	Date	SHA-256	What it said, and why it changed
v0.1	2026-10-02	258c147fe03db3b86ea2b6dc7dee19c064ca	T1 judged as recall of "safety intervention" (≥ 0.95 , ≥ 100 positives in test_unseen); T3 maintenance-needed recall ≥ 0.90 ; T2 and T4 top-1 accuracy ≥ 0.99 ; 40 injected runs per category. The author approved these choices on 2026-10-06

V er si o n	D at e	SHA-256	What it said, and why it changed
		7dff9b e39507 6709a9 5f262e 16d4	
v0 .2	20 26 -1 0- 06	d96cfb 2c41a4 997d13 c9eebc db7b5c d41f71 c4d8b9 cb2273 b4e5c9 6cc3e0 6f0a	A cross-review of v0.1 the same day (three AI judges with different lenses and one AI domain reviewer on PLC maintenance and safety, none reading the others' outputs) found three problems, each reproduced on the pilot labels before adoption. (1) The T1 verdict was decided in advance: first code 18 maps only to safety intervention, so the rule resolved all 69 positive pilot windows and the 39 unresolved windows were all negatives; the composite could only add false alarms. (2) The T3 "maintenance needed" label coincided with a PLC lockout inside the window (17 of 17 pilot windows; 0 of 55 normal or observe windows): cell B has no belt-speed warning, and chatter that passes the filter is itself a lockout (code 9). (3) T2 and T4 were tautological (Section 3.2). v0.2 moved the T1 verdict to multi-class accuracy on rule-unresolved items, made the four risk categories a secondary measure, withheld T3 until the window was fixed, demoted T2 and T4 to consistency checks with ≥ 300 items and 0 dangerous wrong answers, judged on all test items instead of test_unseen alone, and raised the val recall target to 0.97
v0 .3 (f ro ze n)	20 26 -1 0- 06	35b4e3 15a3e0 f6f227 9df408 fd5890 588472 2d0028 99a322 36c0b5 b446d2 fe91	The T3 window was cut before the first PLC warning or lockout, with <code>t3.lead_s</code> and the 5 s rule. The cut alone left 0 maintenance-needed pilot windows, so the degradation catalogue was changed with the label boundaries unchanged: a 2-scan chatter pulse now starts at intensity 0.95 instead of 0.75, and belt-only runs with an even repetition index in their split draw the target deviation from 0.05–0.08. The pilot then had 7 such windows (6 runs). Set size was recomputed with the generator's plan function: 60 injected runs per category (expected about 156 rule-unresolved items) and the wear runs above

All three versions were written before the official set existed. The tool changes behind v0.3 had three AI code-review rounds (1, 0 and 0 required changes). The M1 fixed inputs had three rounds (3, 2 and 1 required changes). The last change round, M1-5 (the fourth after the first version M1-1; Section 6), was committed without a separate review.

4. Pre-registered verdicts and predictions

Pre-registration v0.3, SHA-256 35b4e315a3e0f6f2279df408fd58905884722d002899a32236c0b5b446d2fe91, committed in the private repository (commit 7650685, 2026-10-06 13:53:45 KST) 29 seconds before set generation started. The fixed-input check at scoring time reported "match" (GGUF, model record, prompts, verdict constants, item files, tool hashes and the server's actual arguments).

Verdicts [pre-registered] (test_seen + test_unseen; simulation; physical values assumed; no physical comparison)

Task	Line	Result	Verdict
T1 (cell B, rule-	"Model adds" if $C \geq R0 + 0.10$ with the difference's interval lower bound > 0 ; "rule enough" if $R0 \geq 0.90$ and C does not add; "both below the line" if $R0 < 0.90$ and C does not add. Requires ≥ 100 items and ≥ 30 combination keys	165 items, 141 keys. $R0\ 90/165 = 0.545$ [0.419, 0.669]; $C\ 40/165 = 0.242$ [0.168, 0.331]; $C - R0 = -50/165 = -0.303$ [-0.446, -0.172]	both below the line

Task	Line	Result	Verdict
unresolved)			
T3	"Model adds" if M1 has recall ≥ 0.90 (threshold chosen at val recall 0.97) and a lower false-alarm rate than R0-T3 with the difference's interval lower bound > 0 ; "rule enough" if R0-T3 has recall ≥ 0.90 and false-alarm rate ≤ 0.10 . If both hold, "rule enough" . Requires ≥ 45 positive runs and ≥ 20 per channel (belt, chatter)	53 positive runs (belt 38, chatter 21; a run can count in both), 108 negative runs. R0-T3 recall 51/53 = 0.962 [0.906, 1.000]; false alarms 0/108 (Clopper–Pearson upper 0.034)	rule enough
T2	$R0 \geq 0.99$, 0 dangerous wrong answers, ≥ 300 items	R0 7,323/7,323 = 1.000 (Clopper–Pearson lower 0.9995); 0 dangerous	[PASS]
T4	Same	R0 1,374/1,383 = 0.993 [0.989, 0.997]; 0 dangerous	[PASS]
T1, test_unseen alone	Same as T1	71 rule-unresolved items (< 100)	with held

M1 failed on 0 of the 577 judged items (165 T1 + 412 T3), so the 5% invalidation rule did not apply.

Predictions [pre-registered] (wrong predictions are marked and not revised)

	Prediction	Result	Verdict
1	On rule-resolved T1 items, R0 accuracy ≥ 0.95	700/741 = 0.945 [0.928, 0.960]; 704 were needed	wrong
2	On rule-unresolved T1 items, C is not 0.10 or more above R0	$C - R0 = -0.303$	correct
3	T2 and T4 consistency [PASS]	[PASS], [PASS]	correct
4	R0-T3 recall > 0.90 , so T3 "rule enough"	51/53 = 0.962, "rule enough"	correct

Prediction 4 was close to certain by construction: R0-T3 uses the same boundaries that produced the labels, and the pre-registration said so.

5. Results

The numbers are the scorer's output (score.json), recomputed independently by scripts/check.py (Section 8). Measures named in Sections 6–8 or 10 of the pre-registration are marked [pre-registered]; numbers computed by the hash-fixed scorer or from the public files but not named there are marked [exploratory]. All numbers are from the simulated rig (physical values assumed; no physical comparison).

5.1 T1 lockout cause (cell B, 906 items: 741 rule-resolved, 165 unresolved)

Method	Unresolved (165) [95%] [pre-registered]	All cell B (906) [95%] [exploratory]
R0-full	90 = 0.545 [0.419, 0.669]	790 = 0.872 [0.831, 0.904]
Composite C	40 = 0.242 [0.168, 0.331]	740 = 0.817 [0.775, 0.851]
M1 alone	40 = 0.242 [0.170, 0.327]	349 = 0.385 [0.349, 0.421]
R0-masked (reference)	1 = 0.006 [0.000, 0.020]	255 = 0.281 [0.250, 0.314]
R1	1 = 0.006 [0.000, 0.020]	192 = 0.212 [0.182, 0.242]

On the unresolved items the model's top answer changed with the order of the choices on 148 of 165 (0.90) [pre-registered].

R0-full sees the first fault code; M1 and R0-masked do not. [exploratory] Against the rule that sees the same masked input, M1 was higher (40 vs 1 of 165; 349 vs 255 of 906), but on the unresolved items R0-masked equals the majority baseline R1 (1/165), so beating it shows little.

[exploratory] On the 165 unresolved items, R0's candidate set had three categories in 106 and two in 59, and contained the true cause in 160. The model's top answer lay outside the rule's candidates in 69 items. Had the model's answer been restricted to the rule's candidates (taking the candidate with the highest mean probability), it would have been right in 85 of 165, still below R0's 90. These counts were computed after the results from the public files.

Four risk categories [pre-registered] (cell B; TP / FN / FP; the denominator is the number of positives; recall with 95% interval for each method and category is in `score.json` → `T1.risk4.<category>.<method>.recall`, except brake, below 20 positives)

Category	Positives	R0-full	Composite C	M1 alone
Safety intervention	192	191 / 1 / 0	191 / 1 / 0	89 / 103 / 0
Brake	12 (counts only)	12 / 0 / 0	12 / 0 / 0	0 / 12 / 0
Supply-pressure loss	158	158 / 0 / 41	158 / 0 / 0	156 / 2 / 85
Mechanical jam	54	0 / 54 / 0	34 / 20 / 104	46 / 8 / 347

R0's recall for mechanical jam was 0/54 (Clopper–Pearson upper 0.066) because, among its candidates, mechanical jam was never the most frequent class in train. The composite's 34/54 (0.630 [0.491, 0.755]) comes from the model naming mechanical jam often (347 false alarms when alone), not from evidence that it recognised the cause; overall the composite was below R0. The model alone missed 103 of 192 safety interventions and all 12 brake lockouts. The missed cases are listed by window in `score.json` (`T1.risk4.<category>.<method>.missed`).

By number of injected faults [pre-registered] (cell B, `test_seen`, all items; Section 10 of the pre-registration)

Faults	Items	R0-full	Composite C	M1 alone
1	36	17	13	4
2	59	47	42	22
3	369	325	313	154

Most cell-B `test_seen` items (369 of 464) come from three-fault combinations. `test_unseen` holds only two- and three-fault combinations, so "unseen combination" and "multiple faults" are confounded there.

Confusion tables by method and by split (`test_seen`, `test_unseen`) [pre-registered] are in `score.json` (`T1.confusion`, `T1.confusion_by_split`) and `results/tables.json`; cells with fewer than 20 items are counts only.

Cells A and P [pre-registered, counts only]: A 132 items, R0 132, M1 132; P 228 items, R0 228, M1 3. [exploratory] Both cells have two choices. In cell P the model chose "safety intervention" on 225 of 228 items; in cell A, where every true cause is safety intervention, the same preference gives 132 of 132.

5.2 T3 maintenance sign

412 windows (3 more were removed for a lead time under 5 s); 251 windows were truncated, and the rule and the model saw the same truncated text (0 mismatches). No stroke-time item reached "maintenance needed" before a PLC reaction (Section 7).

Method	Recall (runs) [pre-registered]	False-alarm rate (runs) [pre-registered]	Level accuracy (windows) [exploratory]
R0-T3	51/53 = 0.962 [0.906, 1.000]	0/108 = 0.000 [0.000, 0.034]	391/412 = 0.949 [0.927, 0.968]
M1	53/53 = 1.000 [0.933, 1.000]	108/108 = 1.000 [0.966, 1.000]	161/412 = 0.391 [0.345, 0.437]

R0-T3 uses the label boundaries, so its 51/53 and 0/108 show that the rule and the labels agree; they are not evidence of maintenance skill.

[pre-registered] By channel, R0-T3 caught 37 of 38 belt runs and 20 of 21 chatter runs. In 19 belt-only runs the PLC never reacted to the deviation; these runs exist by design (v0.3 added belt-only runs with a 5–8% target deviation, and cell B has no belt-speed warning). R0-T3 flagged all 19; M1 also flagged all 19, which carries no information because M1 flagged every run. Of the 70 maintenance-needed windows, 4 had a later PLC warning or lockout, 21.2–28.6 s after the window ended; the other 66 had none. Early alarms on negative windows of mixed runs: R0-T3 0, M1 84.

The T3 verdict did not depend on M1. Under the pre-registered tie rule ("rule enough" wins when both "rule enough" and "model adds" hold), the verdict is "rule enough" as soon as R0-T3 meets its line, whatever M1's threshold. M1's threshold was in any case degenerate: it is chosen on val, which held only 2 maintenance-needed runs, so it came out at 0.065 and M1 flagged every run. The set-size change of v0.3 enlarged the test wear runs but left val at one repetition; this is a design defect. Together with the label boundaries in R0-T3, this means the T3 verdict carries no information about the model.

5.3 T2 and T4 consistency

Task	Items	R0 [pre-registered]	R0 dangerous [pre-registered]	M1 alone [pre-registered, reference]	M1 dangerous [pre-registered, reference]
T2	7,323 (all cell B)	7,323 = 1.000	0	1,055 = 0.144 [0.139, 0.149]	6,268
T4	1,383 (A 132, B 1,010, P 241)	1,374 = 0.993 [0.989, 0.997]	0	749 = 0.542 [0.500, 0.582]	100

In T2, every step other than the correct one is one the recovery table cannot reach from the current step, so under this definition every wrong T2 answer is dangerous: M1's 6,268 is all of its wrong T2 answers (7,323 – 1,055).

[exploratory] 228 of the 1,383 T4 items have a single choice (cell A 132, B 73, P 23). R0's fallback to R1 on 132 items is exactly the 132 single-choice items of cell A. On the 1,155 items with two or more choices, R0 was right on 1,146, still above the 0.99 line. M1's 749 also includes the 228 trivial items. R0's 9 T4 errors are all in cell B.

5.4 The model alone [pre-registered, reference, not a verdict]

M1 answered all 10,387 test items (T1 1,266 · T2 7,323 · T3 415 · T4 1,383) with 0 failures, 0 partial failures and 0 missing orders.

Task (all test items)	R0	M1 alone
T1, cell B (906)	790 = 0.872 [exploratory]	349 = 0.385
T2 (7,323)	7,323 = 1.000	1,055 = 0.144
T4 (1,383)	1,374 = 0.993	749 = 0.542
T3 level (412 windows) [exploratory]	391 = 0.949	161 = 0.391

- **Flagged for human review: 8,404 of 10,387 items (81%)** changed their top answer across the orders of their choices. The denominator includes the 228 single-choice items, which cannot

change, and the 658 two-choice items, which had two orders.

- Confident wrong answers (wrong with selection probability ≥ 0.9): 93.
- Selection probability: minimum 0.144, quartiles 0.253 / 0.323 / 0.487, maximum 1.0. Spread across orders: quartiles 0.306 / 0.427 / 0.588, maximum 0.892.
- T1 by true category (M1 alone, cell B): belt drive 0/133 and flow reduction 0/95 correct; sensor 9/119; inspection device 49/143. Summed over categories, mechanical jam was the model's most frequent wrong answer (347), for example 98 of 119 sensor items and 89 of 95 flow-reduction items; for inspection-device items "sensor" was more frequent (61), and for belt-drive items "supply-pressure loss" (72).
- All 7,938 M1 wrong answers are listed in `score.json` (`missed_or_wrong_m1`).

5.5 test_unseen alone [pre-registered]

Tas k	Result
T1	withheld (71 rule-unresolved items, 64 keys). R0 48/71 = 0.676; C 19/71 = 0.268; rule-resolved R0 353/371 = 0.951
T2	R0 3,639/3,639; 0 dangerous
T4	R0 682/684 = 0.997; 0 dangerous
T3	R0-T3 recall 30/32 = 0.938, false alarms 0/49; M1 32/32 and 49/49. Below the count lines (32 runs; belt 19, chatter 19), so it would be withheld on its own

6. Run ledger

Step	Time (KST)	Outcome	Record
Pilot set (development only)	2026-10-02, regenerated 2026-10-06	prompts and choices drafted	not used in results
Last M1 change round, committed without a separate review	2026-10-06, before 13:53	see below	—
Fixed inputs committed	2026-10-06 13:53:23	—	private commit 6 540131
Pre-registration hash committed	13:53:45	—	private commit 7 650685
Official set, generation 1	13:54:14 – 15:28:55	2,997 runs	—
Rule tables from the generation-1 train windows	recorded 15:54:39, committed 15:54:45	before generation 2 finished and before any test file was opened	private commit 0 e0e3f6
Official set, generation 2	started after 15:28:55 (start not logged) – 17:02:52	first comparison: DIFF (below)	—
Generation-1 windows rebuilt at the same commit, no re-simulation; second comparison	17:02:52 – 17:06:41	127,453 files and the manifest byte-identical (SAME); counted as one retry	private commit 1 903508 (17:06:41)
Cross-machine check of 30 official runs (pre-registration E7)	—	not done	the second machine was not available
Items, truncation, rule predictions	from 17:06:44	20,013 items (3 windows skipped)	—

Step	Time (KST)	Outcome	Record
Evaluation script started; M1 in file order	17:06:44 – about 19:29	12,870 result lines (item × order)	stopped
Judged-item lists made; M1 on judged T1 items	from 19:29	1,205 lines (241 rule-unresolved items × 5 orders, val and test)	order changed (below)
Interim T1 numbers	2026-10-06 19:45	computed with the scorer's functions; the verdict numbers equal the final values	—
M1 on judged T3 windows	until 2026-10-07 00:47:52	2,145 lines (429 windows × 5 orders, val and test)	—
Interim T3 numbers	2026-10-07 05:58	all values equal the final ones	—
M1, remaining items	00:47:52 – 15:28:29	three resumed passes, exit 0	—
Official scoring	15:28:39	fixed-input check "match"; verdicts as Section 4	—
Result evidence committed	15:29:33	verdicts and SHA-256 of the large result files	private commit 3 674563
Independent re-run and full ladder replay of the official set	2026-10-08, about 07:37–07:43	re-run 60 runs, 361 windows: 0 mismatches; ladder replay cell B 2,131 runs and packaging 504 runs: first mismatch 0; lab repository state unchanged; positive controls detected	private commit 1 67c85a (07:43:01); results in results/

First determinism comparison. 124,428 of the 127,453 files differed between the two generations. The only difference was the git commit inside the tool-version string in each file header: the rule tables were committed (0e0e3f6) while generation 2 was running. With that string replaced, all files were equal, and the 2,997 raw simulation records were equal from the start. Read literally, the invalidation rule ("the two generations differ") was met. The author's LLM coding agent (the lead session) judged the difference to be the header string only, rebuilt the generation-1 windows at the same commit without re-simulation, and accepted the byte-identical second comparison; this judgement was made after the comparison and before any test item was scored, and it was reported to the author on 2026-10-06, who approved release of this report on 2026-10-08. It is counted as one retry of the same cause (cap: two). After that retry, no invalidation criterion applied. Because of the rebuild, the tool-version string in every released output carries commit 0e0e3f6, not the fixed-input commit 6540131; between the two commits only the three rule tables, their record and the pre-registration hash record changed; the generator tools did not. Item E7 of v0.1 (carried into v0.3) asked for the outputs to carry the fixed commit in their tool-version string; they carry 0e0e3f6 instead, a deviation with no effect on the tools.

Rule tables before generation 2. The rule tables were built from the generation-1 train windows before generation 2 finished, earlier than the pre-registered order (two generations → comparison → rules). Generation 1 was later found byte-identical to generation 2 (after the header rebuild, which changed only header strings).

Execution order and interim numbers. To see the verdict items sooner, the main model run (in file order, from 17:06:44) was stopped at about 19:29, the judged T1 and then T3 items were run, and the main run then resumed. Results are written per (item, order) and appended, and the scorer found no duplicates or gaps. From the judged items, interim T1 numbers (2026-10-06 19:45) and T3 numbers (2026-10-07 05:58) were computed with the scorer's functions, about 20 and 10 hours before official scoring. The numbers used for the verdicts equal the final values; the T1 file's model-alone values over all items differ (for example 40/906 against the final 349/906), because at that time M1 had been run only on the judged items. The T3 file equals the final values throughout. No fixed input changed afterwards (fixed-input check "match").

Independent re-run and ladder replay after scoring. The pre-registration (Section 9) places the independent re-run and the ladder replay of the official set between generation and the rules ("first mismatch 0 to proceed"). They were run only on 2026-10-08, after scoring. Both found 0 mismatches (table above). The order deviates from the pre-registration; had they found a mismatch, the run would have had to be declared invalid after its results were known.

Cross-machine check not done. Item E7, approved in v0.1 and carried into v0.3 by reference ("as approved in v0.1"), asked for 30 runs of the official set to be generated also on a second machine (macOS) and their raw records compared. This was not done because that machine was not available; it is a deviation from the pre-registration. In the pilot set, the raw records of all 357 runs were byte-identical between that machine and the official machine (pre-registration v0.3, Section 11).

Last code change without review. The last change round of the M1 fixed inputs (M1-5) changed how truncated logs are rendered to the model (the remaining lines renumbered from 0, no omission marker), added a cross-check that the rule and the model saw the same text (SHA-256 per item), added tests for the R0-T3 boundaries, and changed `prompts.json`. It was committed without a separate third-party review, which departs from Section 9 of the pre-registration (commit to the repository only after review). It was checked with the previous reviewer's mutation runner: 124 of 125 mutations were caught; the one missed is equivalent in its verdicts.

No invalid or aborted official run after the one retry above.

7. Threats to validity and what this report does not claim

- **The pre-registration was not recorded publicly before the run, and publication was decided after the results.** The pre-registration left publication open, to be decided after the results; the decision to publish was taken on 2026-10-08, after the verdicts were known. Results that are published only after being seen are open to publication bias. For our published reports TR-2026-02, -03, -04 (with its amendment 1), -06 and -07, the pre-registration hash was committed to the public repository (`PREREGISTRATIONS.md`) before the run; for E5 it was committed only to a private repository (commit 7650685, 2026-10-06 13:53:45 KST). A reader can compare the released pre-registration with the hash given here but cannot verify, from a third-party record, that it was fixed before the results.
- It is a simulation. Faults, wear and the operator are simulated, all physical values are assumptions, and no result was compared with physical equipment.
- One model (4B, no additional training), one prompting method (first-letter probability), Korean instructions. Fine-tuned models and other methods were not tested; they are candidates a later version could examine. The result does not rank models.
- **R0-full sees the first PLC fault code; M1 does not.** This asymmetry is part of the design (the composite uses the code where it resolves the cause), so "model alone vs R0-full" is not a like-for-like comparison. R0-masked sees the same input as M1 but on the unresolved items it is no better than the majority baseline.
- **The composite as designed** let the model choose among all eight cell-B categories, not among the rule's candidates. Restricting it to the candidates (85/165, [exploratory], Section 5.1) would not have changed the verdict, but other ways of combining rule and model were not tested.
- **The T1 "rule enough" outcome was practically out of reach.** On unresolved items R0 picks the train-most-frequent of two or three candidates; reaching 0.90 would need one candidate to account for 90% of such items, and such a category would rarely leave the code unresolved. In practice the T1 verdict could only be "model adds" or "both below the line". This is a structural limit of the same kind as the v0.1 problem, found only after the results.
- **R0-T3 uses the label boundaries, and R0 for T2 and T4 is the table that defines the answer.** These checks show that rules and labels agree; they are not evidence that rules are good

maintenance practice. With the tie rule, the T3 verdict carries no information about the model (Section 5.2).

- T3 val was too small: 2 maintenance-needed runs, threshold 0.065, M1 flagged every run.
- Stroke-time channel 0: the PLC reacts at a stroke ratio of 1.138 with a chute-arrival time-out (code 19) before the 1.15 boundary, so no stroke item reached "maintenance needed" before a PLC reaction. The belt-only maintenance-needed runs occur in test_seen only.
- 251 of 412 T3 windows were truncated to fit the 120,000-token limit; in the official test set the truncated T3 items kept between 126 and 203 s of the 300 s window (median 199 s), estimated as the share of log lines kept $\times$ 300 s, assuming lines are spread evenly in time. The rule read the same text.
- Cumulative counters in the log (for example the belt count input) reveal how long the run has been going. Truncated and full windows share one format, but elapsed time can still be computed. For an untrained model the risk is low; it is not measured.
- Cells A and P were not judged; there the cell alone fixes the cause, and the model's numbers there reflect a preference for "safety intervention" (Section 5.1).
- Catalogue gaps named by the AI domain reviewer of Section 3.6: no cylinder position-switch, valve or safety-device faults, no gradual supply-pressure or brake wear, no return-stroke trend, no I/O or power faults; "safety intervention" covers only safety functions that worked. Lockouts that also occur in the fault-free baseline run are removed from T1; this includes code-19 lockouts caused by the simulated operator not emptying the chute. Code-19 lockouts with an injected root cause remain (46 of the 165 unresolved items have first code 19). The packaging runner's reset and start wiring differs from its specification.
- Run-order deviations from the pre-registration (Section 6): rule tables built before generation 2 finished; the first determinism comparison failed on a header string and was resolved by one retry; judged items run first, with interim numbers computed before official scoring; the independent re-run and ladder replay done after scoring; the cross-machine check (E7) not done; the outputs carry commit 0e0e3f6 rather than the fixed commit (E7); the last code change committed without review.
- The pre-registration (its Section 3) asked for the removed windows to be counted by split and category; only counts by kind are reported (Section 3.1).
- Three released rule tables differ in SHA-256 from the pre-registration's Section 12 table because Section 12-③ required them to be rebuilt from the official train split (Section 3.5); a reader comparing only the Section 12 table will see a mismatch.
- The frozen v0.3 still carries a status row saying "draft, no official set generated" and a "next steps" list that predates the freeze. These are inconsistencies inside the frozen document, recorded here rather than edited.
- The pre-registration was not written for publication: it contains internal approval notes, private repository names and working paths. It is released unchanged because its hash is fixed.
- The 2,997 runs come from one generator and one catalogue; items within a combination key are not independent, so intervals are clustered by key or by run and no significance test is made.
- A regression test of the data tools (phase5a_belt_default) fails on this machine because one gzip header byte (operating system) differs; the decompressed content is equal.
- All reviews (pre-registration, cross-review, code reviews, the review of this report) were done by AI agents of the same family as the agents that wrote the code. No person reviewed the generator, rules, runner or scorer line by line.

8. Released files, and what a reader can and cannot check with them

The report (TR-2026-08.md, TR-2026-08.pdf) sits at the root of the public repository, outside the release bundle. The bundle TR-2026-08/ has 38 files plus MANIFEST.md, which lists each with SHA-256, size and

licence. The only file over 5 MB, `results/ml_perms.jsonl.gz`, is in the Zenodo record only, so the repository copy has 37 files plus `MANIFEST.md`.

File	Content	Licence
<code>README.md</code>	Bundle guide, how to rerun the checks, scope	CC BY 4.0
<code>MANIFEST.md</code>	SHA-256, size and licence of every bundle file; sensitive-string search	CC BY 4.0
<code>LICENSES/</code>	CC BY 4.0 and MIT licence texts	CC BY 4.0
<code>fixed/E5_사전등록_v0.3_2026-10-06.md</code>	Frozen pre-registration v0.3 (Korean), byte-identical; SHA-256 35b4e315...46d2fe91	CC BY 4.0
<code>fixed/history/</code>	Pre-registration v0.1 and v0.2 (Korean), byte-identical (SHA-256 in Section 3.6); design history only, not fixed inputs of v0.3	CC BY 4.0
<code>fixed/e5/</code>	14 of the 26 fixed inputs, byte-identical to the files used in the official run: catalogues (cause classes, faults, degradation, calibration, held-out combinations), official set settings, verdict constants (<code>judge.json</code>), model record, prompts and orders, the four rule tables, T4 priority table. Three rule tables carry the rebuilt hashes (Section 3.5)	CC BY 4.0
<code>fixed/tools/e5_score.py</code>	The official scorer as run (hash-fixed); for reference — the check does not use it	MIT
<code>results/items.jsonl.gz</code>	20,013 items of all splits: ids, task, split, cell, combination key, run, answer, choices and task metadata; no model input text (the SHA-256 of each item's text instead)	CC BY 4.0
<code>results/rules_pred.jsonl.gz</code>	All rule predictions (R0-full, R0-masked, R0-T3, R0 for T2 and T4, R1), the official file byte-identical inside the gzip	CC BY 4.0
<code>results/ml_perms.jsonl.gz</code>	M1 per item and order (94,542 lines): probabilities in the original choice order, validity flags. 9.2 MB: in the Zenodo record only	CC BY 4.0
<code>results/score.json · score.md · truncation.jsonl</code>	Official scorer output and T3 truncation record, byte-identical	CC BY 4.0
<code>results/provenance.json</code>	SHA-256 and size of the original result files; M1 run header; rule-table record	CC BY 4.0
<code>results/evidence/</code>	Five records from the private repository: pre-registration hash record, rule-table record (one absolute path replaced), two-generation determinism record, independent re-run and ladder replay of the official set (2026-10-08); the others byte-identical	CC BY 4.0
<code>results/check_output.json · check_output_repo.json · tables.json</code>	Outputs of the scripts below (full mode, repository mode)	CC BY 4.0
<code>scripts/check.py</code>	Independent recomputation of every verdict, numerator, denominator and interval from <code>results/</code> and <code>fixed/</code> (standard library and numpy; does not import the scorer). Two modes, chosen by	MIT

File	Content	Licence
	whether <code>m1_perms.jsonl.gz</code> is present: the full mode (Zenodo bundle) recomputes everything; the repository mode recomputes the rule baselines, the T2, T3 and T4 verdicts and the fixed-input hashes, and marks the 92 values that need the model's per-order probabilities (including the T1 verdict) as not run	
<code>scripts/table_s.py</code>	Section 5 tables; some are pre-registered, some exploratory (Section 5 marks)	MIT

Of the 26 fixed inputs, 15 are released (the 14 under `fixed/e5/` and the scorer). The other 11 are not: three label and input schemas, two pilot files (pilot set settings and pilot manifest), the record-noise catalogue, the T4 priority rule document (Korean, internal), and the prompt builder, rule code, model runner and their test. The pre-registration itself is not one of the 26; it is released.

The pre-registrations (v0.3 and the history versions) and the evidence records are released unchanged, as in TR-2026-06: they contain internal approval notes, private repository and working-folder names and names of candidate models for later versions. The patent application number and claim summary that appear in the pre-registration's patent check were already published in the TR-2026-04 pre-registration v0.4, Section 13; this report does not disclose them for the first time.

Also not released: the model input texts (PLC log texts), the simulated data set and its generator, the PLC programs and the PLC simulator, the item builder, the model weights (public upstream; only commit and GGUF SHA-256 are given), the run logs and the private repository.

Independent check. In full mode, `scripts/check.py` agrees with `score.json` on all 1,384 verdicts, numerators, denominators and list entries and on all 562 interval ends, rates, the threshold and quantiles. In repository mode it agrees on the 531 and 216 values it can recompute and marks 92 as not run. Both modes exit with code 0; `scripts/tables.py` needs the full bundle. The bootstrap uses the pre-registered seed string with the same per-measure naming as the scorer, so intervals can be compared digit by digit; the counting logic was written separately.

A reader **can**, with the Zenodo release bundle: recompute every verdict, numerator, denominator and interval in Sections 4 and 5, M1's answers, the human-review flags and the T3 threshold; recount the tables; compare the released pre-registrations and fixed inputs with their SHA-256. With the repository copy alone (no `m1_perms.jsonl.gz`), the repository mode of `scripts/check.py` recomputes the rule results and the T2, T3 and T4 verdicts (the T3 verdict follows from R0-T3 alone under the tie rule), but not the T1 verdict or any model value. A reader **cannot**: rerun the model (the input texts are not released); re-simulate the runs (the generator and the PLC simulator are not released); check the 11 unreleased fixed inputs; verify from a third-party record that the pre-registration preceded the results, that the rule tables were recorded before test was opened, or the two-generation comparison and its times (the records in `results/evidence/` are copies from the private repository, whose history is not public); confirm that the files came from the stated code.

Licence and notices

Code (`.py`) is under the MIT licence; everything else in the bundle and this report is under CC BY 4.0. This release does not grant any licence under patents of WACE Inc., including Korean patents No. 10-2904772 and No. 10-3003470, or under any pending or future patent application of WACE Inc. Qwen is a model of the Qwen team (Alibaba Cloud); this report is not affiliated with them and the release contains none of their files. NVIDIA, CUDA and GB10 (named in the pre-registration as the machine used) are trademarks of NVIDIA Corporation; this report is not affiliated with NVIDIA. Other product names are trademarks of their owners.

9. Checks before release

Gate	Status
Report number TR-2026-08 is free: not in the public repository waceinc/tech-reports or on the internal NAS (2026-10-08)	[PASS]
① Pre-registration hash: the released v0.3 equals the file whose SHA-256 (35b4e315...46d2fe91) is in private commit 7650685. A public commit before the run does not exist for this report; this replaces the usual gate and is stated in the abstract and Section 7. The bundle's v0.3 equals the recorded value; re-checked independently by the final review	[PASS]
② Invalidation criteria: fixed-input check "match"; the two generations byte-identical after one retry; M1 failures 0 (line: 5%); independent re-run of the official set (60 runs, 361 windows) and full ladder replay (2,131 + 504 runs) with 0 mismatches — run after scoring (Sections 6 and 7)	[PASS]
③ Independent check: scripts/check.py reproduces the scorer output — full mode 1,384 verdicts, numerators and denominators and 562 interval values equal to score.json; repository mode 531 and 216 equal, 92 not run	[PASS]
④ Third-party review by AI agents, given the text as an outside draft, until the final review has 0 required changes: the final review of v0.5 had 1, applied in this version	[PASS]
⑤ Fact check of company and person statements: the author's name, affiliation and e-mail, the company's location, its two patent numbers (as in TR-2026-07) and the earlier reports' public pre-registrations (as in PREREGISTRATION S.md); internal approval notes inside the hash-fixed pre-registrations are released unchanged	[PASS]
⑥ Number check against the result files: v0.2 and v0.3 checked value by value; the 3 + 3 discrepancies found are corrected, and the re-review recounted 32 values from the public files with 0 mismatches.	[PASS]
⑦ Recomputation with private paths blocked: on 2026-10-08 the final bundle was copied and scripts/check.py was run in both modes under a Python audit hook that refuses to open any file outside the bundle and the Python libraries (positive control: reads of a private result file and of a private work-folder file were refused). Both outputs were regenerated and are byte-identical to the released ones (full mode 1,384 values and 562 intervals; repository mode 531 and 216). Earlier, the v0.3 number check, the v0.3 re-review and the final review recomputed separate copies in both modes without blocking, with the same outputs	[PASS]
⑧ Sensitive strings 0 in all files, inside archives and gzip headers, with a positive control; internal names in the hash-fixed pre-registrations and evidence records are released unchanged by decision (Section 8)	[PASS]
Forbidden-term sweep (company term list, run on v1.0 on 2026-10-08: 0 hits)	[PASS]
⑨ MANIFEST: every file's SHA-256 matches (38/38 after the README update)	[PASS]
⑩ Patent keyword comparison with positive control, repeated on 2026-10-08 on the whole bundle (archives decompressed) and on report draft v0.4 (later versions changed only the front matter, the licence notice and Section 9): report 0; bundle only the patent-check passage inside pre-registration v0.3 (the same content as the already published TR-2026-04 pre-registration v0.4, Section 13) and "RAG" inside "paragraph" in the CC licence text (false positive)	[PASS]
Author's release confirmation	given 2026-10-08

10. Use of generative AI

The fault generator and labelling tools, the rule baseline, the model runner, the scorer, the pre-registration, an internal summary of the results (not released) and this report were written with LLM coding agents. The third-party review of the pre-registration (2026-10-02), the cross-review of three independent judges and one domain reviewer (2026-10-06), the code reviews of the T3 window change and of the M1 fixed inputs, the review of draft v0.2 of this report, its number checks, the re-review of

draft v0.3 and the final review of draft v0.5 were also done by agents of the same family. The independent judges of the cross-review did not read each other's outputs, and the code reviewers did not read the author agent's reports. Re-reviews read the previous review. The review of draft v0.2 and its number check did not read each other; the v0.2 review cited the author side's bundle and result reports, and the re-review of v0.3 read the previous review and the second bundle report. The author decided the open items, approved the design changes, the frozen pre-registration and the commits, and decided the release after the results.

Appendix A. Where each number comes from

Keys are those of `results/score.json` (category names in keys are in Korean: 안전 개입 safety intervention, 브레이크 brake, 공급압 저하 supply-pressure loss, 기계 걸림 mechanical jam). `check_output.json` is the output of `scripts/check.py` on the released files. Sources marked (private) are in the private repository or run logs and are not released.

Number	Source
Four verdicts (Abstract, Section 4)	<code>score.json</code> → verdicts; recomputed in <code>check_output.json</code>
T1 unresolved: 165 items, 141 keys; R0 90, C 40, M1 40, R1 1, R0-masked 1, intervals	<code>T1.n_undetermined</code> , <code>T1.undetermined_keys</code> , <code>T1.undetermined_acc</code> . {R0, C, M1, R1, R0-가림}
$C - R0 = -50/165 = -0.303$ [−0.446, −0.172]	<code>T1.diff_C_minus_R0</code>
Prediction 1: 700/741 = 0.945	<code>T1.determined_acc_R0</code>
Cell B all: 790, 740, 349, 255, 192 of 906	<code>T1.all_acc.*</code> , <code>T1.n_cell_b</code>
Human check 148/165	<code>T1.human_check.undetermined</code>
Candidates 3/2 (106/59), truth in candidates 160, model outside candidates 69, restricted 85/165	counted from <code>items.jsonl.gz</code> , <code>rules_pred.jsonl.gz</code> , <code>m1_perms.jsonl.gz</code> (exploratory)
Four risk categories (TP/FN/FP, recall, missed lists)	<code>T1.risk4.<category></code> . {R0, C, M1}
By number of faults (36/59/369)	<code>T1.test_seen_by_faults</code>
T1 by true category, confusions, by split	<code>T1.confusion</code> . {C, M1}, <code>T1.confusion_by_split</code>
Cells A and P	<code>T1_counts_AP</code>
T3: 412 windows, 3 excluded, 53/108 runs, channels 38/21/0	<code>T3.n_windows</code> , <code>T3.excluded_short_lead</code> , <code>T3.n_pos_runs</code> , <code>T3.n_neg_runs</code> , <code>T3.channel_pos_runs</code>
T3 recall, false alarms, level accuracy	<code>T3.r0.recall</code> , <code>T3.r0.fpr</code> , <code>T3.m1.recall</code> , <code>T3.m1.fpr</code> , <code>T3.r0_level_acc</code> , <code>T3.m1_level_acc</code>
T3 by channel 37/38, 20/21	<code>T3.r0_recall_belt_speed_dev</code> , <code>T3.r0_recall_chatter</code>
Threshold 0.065, 2 val positive runs	<code>T3.threshold</code> , <code>T3.threshold_note</code>
Belt-only, PLC unaware: 19, 19, 19	<code>T3.belt_solo_plc_unaware</code>
Lead times 21.2–28.6 s, 4 of 70	<code>T3.lead_s_maint</code>
Truncated 251, mismatches 0; early alarms 0 / 84	<code>T3.truncated_windows</code> , <code>T3.truncation_mismatch</code> , <code>T3.early_alarms</code>
T2: 7,323, R0, dangerous, M1	<code>T2.n</code> , <code>T2.r0</code> , <code>T2.r0_dangerous</code> , <code>T2.m1</code> , <code>T2.m1_dangerous</code>
T4: 1,383, R0 1,374, fallback 132, by cell, M1	<code>T4.n</code> , <code>T4.r0</code> , <code>T4.r0_dangerous</code> , <code>T4.r0_fallback</code> , <code>T4.r0_by_cell</code> , <code>T4.m1</code> , <code>T4.m1_dangerous</code>
T4 single-choice 228 (132/73/23); 1,146/1,155	counted from <code>items.jsonl.gz</code> , <code>rules_pred.jsonl.gz</code> (exploratory)
Orders per item 9,501 / 658 / 228	counted from <code>m1_perms.jsonl.gz</code>

Number	Source
M1: 10,387 items, 577 judged, 0 failures, human check 8,404 (81%), 93 confident wrong, quantiles	m1_status.* (items_with_m1, judged_items, judged_failed, partial_failed, missing_perms, human_check, confident_wrong, prob_quantiles, spread_quantiles)
M1 wrong answers 7,938	missed_or_wrong_m1
test_unseen (Section 5.5)	test_unseen.*, T1_test_unseen
Fixed-input check "match"	header of score.md
Rule-table hashes and record time	results/evidence/rules_record_official.json, results/provenance.json; check_output.json → fixed_hash_check
Official re-run 60 runs / 361 windows; ladder 2,131 / 504 runs	results/evidence/official_verify.json, official_ladder_full.json (private commit 167c85a)
2,997 runs, 127,453 files, first comparison 124,428	results/evidence/official_determinism.json
20,016 windows; noise variants 75,405	set manifest (private)
Removed windows by kind (Section 3.1)	set manifest → excluded_windows (private)
Truncated T3 test items kept 126–203 s (median 199 s)	truncation.jsonl (lines dropped) and items.jsonl.gz (T3 body lines), 254 truncated test items
M1 run order: 12,870 / 1,205 / 2,145 lines	order of lines in the original M1 result file and times of the judged-item lists (private)
Pre-registration SHA-256 and fixed-input commit	results/evidence/preregistration.json; SHA-256 recomputable from fixed/
Run times, interim computations and commits (Section 6)	run logs and the private repository's git log (private)